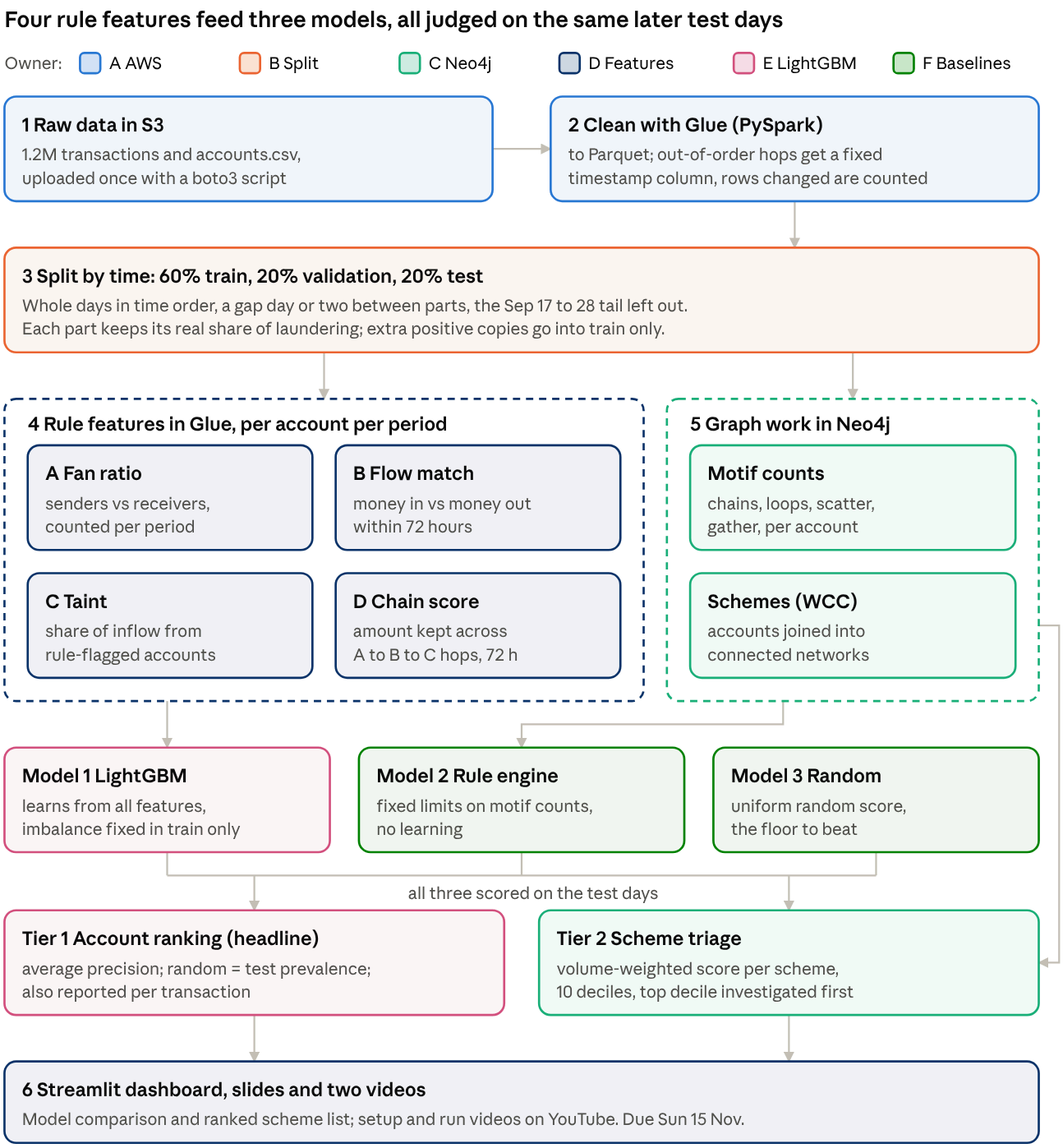
Q1 plan and work split

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The plan at a glance

We keep all 1.2M transactions, split them by time, turn each account's behaviour into rule scores, and test whether LightGBM ranks laundering accounts better than fixed rules and a random guess. The best model's scores are then rolled up into whole schemes so an investigator knows which network to look at first.



Q1 pipeline end to end · 6 stages, 3 models, 2 scoring tiers

Read it top down. Steps 2, 4 and 5 hold the long jobs (3 to 4 hours each), and 4 and 5 run side by side. Box colour shows which member owns that step.

Your steps

Until we meet on Wednesday, everyone works on their own, at their own pace. There are no deadlines. The only aim is to have something to show at the meetup.

Pick a letter in the group chat, then follow only your list. Each list starts with what you can do straight away, then what has to wait for someone else. When you finish a step marked **post**, post it in the group chat, because someone is waiting on it. While you wait, do your "start now" steps. The fixes from the risk table below are already written into these steps.

Letter	Lane	Waits for
A	AWS and data cleaning	nobody
B	Split dates and labels	A's AWS keys
C	Graph work in Neo4j	B's dates, for the long run
D	Rule features, then the dashboard	A's clean data and B's dates
E	LightGBM model	D's features
F	Baselines, the oversampling check, slides	D's features and C's motif counts

A: AWS and data cleaning

Before Wednesday

- [ ] Start now: get working AWS keys and check that one Athena query runs. **Post** when it works, because everyone else needs AWS.
- [ ] Then start the Glue job that cleans all 1.2M rows to Parquet. It takes 3 to 4 hours, so just let it run. Add a corrected timestamp column, keep the original one, and note how many rows changed.
- [ ] When it finishes, **post** where the clean data is saved.

After the meetup

- [ ] Write one script that runs the whole pipeline from start to end.
- [ ] Record the two videos with B (set up the pipeline, then run it) and upload them to YouTube.

B: split dates and labels

Before Wednesday

- [ ] Start now: find out which file the 1.2M rows come from, and read the split design doc.
- [ ] Once A posts the keys, run split queries 1 to 3, then `choose_cuts.py`.
- [ ] **Post** the train, validation and test dates. C, D and E wait on these.
- [ ] Build the split table, and mark each account illicit or not for each period. **Post** the rule you used.

After the meetup

- [ ] Draw the architecture diagram for the slides.
- [ ] Record the videos with A.

C: graph work in Neo4j

Before Wednesday

- [ ] Start now: load the accounts and transactions into Neo4j, keeping the timestamps (1 to 2 hours). The raw CSV is fine for this.
- [ ] Once B posts the dates, start the motif count for each account in each period: chains, loops, scatter, gather (3 to 4 hours). Run WCC on the test days in the same session.
- [ ] **Post** the motif counts for F, and how big the biggest WCC groups are. If one group holds most accounts, run WCC again on rule-flagged transactions only.

After the meetup

- [ ] Use E′s scores to give each scheme a risk score, then split the schemes into 10 deciles.

D: rule features, then the dashboard

Before Wednesday

- [ ] Start now: write Rules A to D in PySpark and test them on one day of raw data. Follow 4 rules: one row per account per period; only data up to the end of that period; seed Rule C taint from rule hits, not the laundering labels; in Rule D, pair transfers only within 72 hours.
- [ ] Once A′s clean data and B′s dates are posted, start the full feature run (3 to 4 hours).
- [ ] **Post** where the feature table is saved, for E and F.

After the meetup

- [ ] Add a Q1 page to the Streamlit dashboard with the model comparison and the ranked list of schemes.

E: LightGBM model

Before Wednesday

- [ ] Start now: change `train_eval.py` to score accounts and print average precision and the decile lift. Test it on fake data, locally or in Colab.
- [ ] Once D posts the features, do a first training run and bring the validation score to the meetup.

After the meetup

- [ ] Train the final model and score the test period, with and without Taint, per account and per transaction. **Post** the scores for C and F.

F: baselines, the oversampling check, slides

Before Wednesday

- [ ] Start now: write Model 3 (a random score per account) and the rule-limit logic for Model 2, and test them on fake data.
- [ ] Once D posts the features, run the oversampling pilot twice: once copying positives into train only, once into validation and test as well. Bring both scores. If the features aren't ready, bring the fake-data result.

After the meetup

- [ ] Score Model 2 (on C's motif counts) and Model 3 on the test period.
- [ ] Make the results slide comparing all three models, with the test laundering rate beside every score.
- [ ] Put everyone's slides into one deck.

Everyone

- [ ] **Wednesday, 6 PM in the lab:** show what you have, settle the open questions at the bottom of this doc, and agree the next steps.
- [ ] **After the meetup:** freeze the results, then do the videos, slides, a 20-minute rehearsal and the eLearn submission before the project is due.

Fix these before the long jobs run

Four parts of the plan let the model see the answer or the future, so its scores would look better than they really are. Each fix changes how a long job is written, so agree them in the group chat before Monday's runs. Rows 5 to 7 are smaller issues found while reading the plan.

#	Risk	Why it hurts	Fix	Who
1	Rule C taint is seeded from the laundering labels	Any account paid by a labelled launderer gets taint above 0, so the feature is close to the answer itself. On test it uses test labels, which a bank would never have.	Seed taint from label-free rule hits (Rules A, B, D, as Q1's `v_detect_full` does), or from train-period labels only, spreading forward in time. Report AP with and without Taint.	D, E
2	Oversampling the validation and test sets	Copied positives raise prevalence, and AP rises with prevalence. The score then says nothing about real traffic, and repeated copies are easy hits.	Oversample or class-weight the train set only. If val or test has too few positives, move the day cuts (`choose_cuts.py --min-pos`) instead. If the group keeps the prof's version, report both, real prevalence first.	F, B
3	Features are per account, but the split is by time	An account active in all three periods gets one feature row from its whole history, so its train row already knows what it did in the test period.	One row per account per period. Features use only transactions up to the end of that period. Label = a laundering transaction inside that period. Also report AP on accounts first seen in test.	D, B

#	Risk	Why it hurts	Fix	Who
4	Rule A degrees are counted over all 1.2M rows	In-degree and out-degree include counterparties from the future.	Count unique counterparties per period (or in a trailing window), the same way as row 3.	D
5	WCC on the full graph may give one giant component	Big merchant hubs can link most accounts together, leaving one huge "scheme" and crumbs, so the deciles mean nothing.	C posts the component sizes before Wednesday. If one component holds most accounts, run WCC only on rule-flagged edges, or on top-scored accounts, in the test period.	C
6	Prof said the main evaluation is per transaction; the plan's headline is per account	The marker may expect a transaction-level number.	Report both. Transaction score = the higher of its sender's and receiver's scores, one extra query.	E
7	Rule D pairs blow up at hub accounts	An account with 1,000 transfers in and 1,000 out has a million in/out pairs.	Pair an outbound transfer only with inbound ones in the 72 hours before it (Rule B's window). Roll up to the account as the max score. Use min/max like Rules A and B so the score stays in 0 to 100.	D

Open questions for the meetup

The first two need the prof. If F's oversampling pilot shows a big gap, book her for a short check after the meetup.

- [ ] Did she mean oversampling validation and test, or only training? The consult notes are auto-generated and may have it wrong.
- [ ] Is the headline per account (the plan) or per transaction (what she said)? Default: report both.

- [ ] Which file is the 1.2M rows? The split design and the pilot used HI-Medium (32M rows). The cut search and the Sep 17 to 28 tail only carry over if it is the same data. B checks this first.
- [ ] What makes an account illicit: any laundering transaction sent or received in the period, or sent only? B posts the rule so D, E and F all use it.
- [ ] Drop the Sep 17 to 28 tail (about 58% laundering, a generator artifact)? Recommended: yes, and say so in the write-up.
- [ ] Who presents which part? Default: each member presents their own lane.